Weekly Report 7: Diamond Carving with Larger Viewing Angles

Project Number: 18

1 Introduction

Following our previous experiment on diamond carving with small angular increments, this week we focused on analyzing and preparing for **diamond carving with larger viewing angles**. The purpose of this study is to understand how increasing the angular gap between consecutive silhouettes affects the quality and completeness of the reconstructed 3D model. While the 1.8° setup ensures dense view coverage and smooth model formation, larger angles are expected to reduce computational effort but may lead to less detailed reconstructions.

As the actual diamond silhouette dataset from the collaborating company is yet to be received, this week's work primarily involved theoretical comparison, experimental planning, and setting up the framework for the upcoming tests.

2 Methodology

The carving process remains based on the previously developed visual hull approach, where a voxel grid is refined by projecting voxels onto 2D silhouettes and removing those outside the object boundary. The key difference lies in the **angular spacing between views**: instead of a small angle increment, we will use larger angles (e.g., 10°, 15°, or 20°) to simulate sparser viewpoints.

2.1 Conceptual Steps

- Create or import voxel grid as the 3D space representation.
- Use silhouette images of the diamond captured from widely spaced angles.
- For each view, project the voxel grid onto the 2D plane corresponding to the camera position.
- Carve away voxels that fall outside the silhouette region.
- Combine results across all views to form the final 3D shape.

This adjustment in angular sampling will help us observe the trade-off between accuracy and efficiency in 3D reconstruction.

3 Expected Outcomes

From theoretical understanding, we anticipate the following:

- **Reduced computational load** due to fewer silhouette images and projections.
- **Less smooth and less complete 3D surfaces**, as larger angular gaps leave unobserved regions between views.
- **Sharper geometric outlines** might still be preserved for structured shapes like diamonds.
- The 3D model reconstructed with larger angles is expected to appear slightly *coarser* or *incomplete* compared to the 1.8° reconstruction.

4 Discussion

This phase is crucial for understanding the balance between the number of viewpoints and reconstruction quality. Smaller angular increments capture more details but require significant processing time and memory, whereas larger angles speed up computation but may cause visible gaps or unevenness in the final model.

Once the company dataset is obtained, both small-angle and large-angle carving results will be compared quantitatively and visually to determine the optimal angular spacing for efficient yet accurate 3D reconstruction.

5 Conclusion

This week's progress included:

- Designing the experiment for diamond carving using larger viewing angles.
- Studying the theoretical difference between 1.8° and larger-angle reconstructions.
- Predicting the effects on the reconstructed model's smoothness and completeness.

The next step will involve applying this methodology to the real diamond silhouettes as soon as the dataset is received, enabling practical evaluation of the reconstruction quality at different angular intervals.